

Cancers One by One

Lung, bladder, pancreas, kidney, thyroid, lymphomas, leukemias, stomach, esophagus, testis, brain and non-melanoma skin: genes, scores and organ prevention

Sérgio F · 03-10-2026 · local time

Informative and educational genetic report, based only on Sérgio F genes (genotyping chip + local imputation). It is neither a diagnosis nor a prescription. All actions are marked as suggestion (not applied).

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1. Summary

Favourable. Several organs are clearly below average. In lymphomas: non-Hodgkin in percentile 1.1, diffuse of large B cells in 5.4, follicular in 0.3 and Hodgkin in 0.8; the HLA class II does not have the risk allele. In kidney: 0.8 and 22.4; CCND1 and SCARB1 without risk alleles. In the non-melanoma skin: basal cell 13.0 and 11.2, squamous cell 3.4 and 2.4 MC1R without the variants R, IRF4 C/C. In the testis: 23.9 and 7.9; KITLG A/A without the risk allele. In the stomach: 16.4 and 6.3, both of low reliability; MUC1 C/C and PLCE1 A/A. **TP53 rs78378222** and **BRCA2 K3326X** are normal.

Unfavourable. The bladder is the most charged organ: 94.9, 90.4 and 86.1 scores (the genomic is in 57.2). We join NAT2 slow acetylator (A/A), **PSCA T/T**, **8q24 T/T** and **UGT1A A/A**, as in §83. The glioma is in percentile 90.8 (glioblastoma in 63.8), with **TERT rs10069690 T/T** and **RTEL1 G/G**. The pancreas is in the 83.1 in the 22 variant score (the genomic in 45.0), with NR5A2 A/A. The esophagus is in 84.8 (the genomic adenocarcinoma in 45.3). The thyroid is in 79.1 and 80.7 (detail in §108). Lymphocytic leukemias are above the median but typical: LLC in 68.9 and 78.4, LLA in 73.1.

What does it mean? The map by organ is heterogeneous and no risk is monogenic. The high signs (bexiga, glioma, pancreas in the small score, esophagus and thyroid) are approximately in the quintile or upper decile. The genomic scores of the pancreas and bladder are in the middle, so the real risk is probably moderate. The lung is typical (23.2, 62.6, 50.1), and the CHRNA5 and TERT rs2736100 only weigh with tobacco. To close the map: colorectal cancer is in §102 (95 variants in percentile 94.3), prostate in §58 (very unlikely) and melanoma in §24 (scores in the 6.5 and 25 percentiles).

2. How to read this report

- **Data:** Sérgio F chip genotypes (GRCh38, strand +) and, when SNP is not in the chip, locally imputed genotypes (Beagle 5.5, public panel of genomes complete to 30x) with quality $DR2 \geq 0.8$. Under this threshold SNP is treated as SNP not found in the data.
- **Keyvariants** 40 Selected SNPs from the literature (GWAS Catalog, ClinVar, dbSNP and reference studies): 39 of the chip, 1 imputed and 0 not found.
- **Strand:** All genotypes are in the GRCh38 strand +. The published effect allele is compared with the observed alleles; 4 SNPs needed conversion to the complementary strand and 4 are palindromic (A/T or C/G), solved by the reference allele and the position. When the literature uses the strand of the gene, the letters may differ from those of this report.
- **Polygenic scores (PGS):** 33 PGS Catalog scores calculated locally with 10.258.262 variants in total (chip + imputed). The percentile compares Sérgio F with the reference population ($n = 633$) calculated with exactly the same variants. Percentage ≥ 80 : MAIS PROVÁVEL/ALTO; ≤ 20 : MENOS PROVÁVEL/BAIXO; between 20 and 80: TÍPICO.
- **Judgment:** Each result is divided into Favourable or Unfavourable (or typical/without judgment). A genotype is a probability, not a destination; almost all effects are small and add to the lifestyle.
- **Evidence:** Strong : replicated and with clear clinical effect; Moderate = GWAS/consistent meta-analysis, modest effect; Weakness : small or mechanistic studies.

3. Polygenic scores

In all these scores, a higher percentile = more risk. When there is more than one score per organ, the genomics (centens of thousands to millions of variants) better capture the risk, and the small ones reflect mainly the strong loci. The values already used in §24, §75, §83, §94 and §108 remain.

Lung

Trait	Score (PGS Catalog)	Variants used	Percentile	Result	Judgement
Lung cancer	PGS000078 (Graff 2021, 109 variants)	98 109 (weighted coverage 90%)	23,2	TIPIC	typical
Lung cancer (genomic)	PGS004691 (Hu 2024, LDpred2, 950 thousand variants)	909.458 950.353 (weighted coverage 96%)	62,6	TIPIC	typical
Lung cancer (genomic)	PGS005397 (Boumtje 2026, LDpred2, 1.02 million variants)	992.658 1.022.277 (weighted coverage 96%)	50,1	TIPIC	typical

- **Lung cancer:** Já no §24.
- **Lung cancer (genomic):** In §92 (Hypoxy).

Bexiga

Trait	Score (PGS Catalog)	Variants used	Percentile	Result	Judgement
Bladder cancer	PGS000782 (Kachuri 2020, 15 variants)	14 15 (weighted coverage 94%)	94,9	MAIS PROVABLE	unfavourable
Bladder cancer	PGS000071 (Graff 2021, 15 variants)	14 15 (weighted coverage 95%)	90,4	MAIS PROVABLE	unfavourable
Bladder cancer	PGS012553 (Kothers 2023, 24 variants)	23 24 (weighted coverage 95%)	86,1	MAIS PROVABLE	unfavourable
Bladder cancer (genomic)	PGS004687 (Hu 2024, 1.08 million variants)	1.051.651 1.077.775 (weighted coverage 97%)	57,2	TIPIC	typical

- **Bladder cancer:** In §83 (Bexiga).
- **Bladder cancer:** Já no §24.
- **Bladder cancer:** Já no §83.
- **Bladder cancer (genomic):** Já no §83.

Pancreas

Trait	Score (PGS Catalog)	Variants used	Percentile	Result	Judgement
Pancreatic cancer	PGS000083 (Graff 2021, 22 variants)	21 22 (weighted coverage 97%)	83,1	MAIS PROVABLE	unfavourable
Pancreatic cancer (genomic)	PGS004693 (Hu 2024, AnnoPred, 6.35 million variants)	5.963.031 6.351.686 (weighted coverage 89%)	45,0	TIPIC	typical

- **Pancreatic cancer:** Já no §24.
- **Pancreatic cancer (genomic):** In §55 (Pancreas).

Rim

Trait	Score (PGS Catalog)	Variants used	Percentile	Result	Judgement
Kidney cancer	PGS000076 (Graff 2021, 19 variants)	16 19 (weighted coverage 86%)	0,8	MENOS PROVABLE	favourable
Kidney cancer	PGS004908 (Purdue 2024, 107 variants)	100 107 (weighted coverage 93%)	22,4	TIPIC	typical

- **Kidney cancer:** Já no §24.

Tyroid

Trait	Score (PGS Catalog)	Variants used	Percentile	Result	Judgement
Thyroid cancer	PGS000087 (Graff 2021, 12 variants)	12 12 (weighted coverage 100%)	79,1	TIPIC	typical

Trait	Score (PGS Catalog)	Variants used	Percentile	Result	Judgement
Thyroid carcinoma (genomic)	PGS005260 (White 2025, PRS-CS, 1.09 million variants)	1.063.714 1.085.170 (weighted coverage 97%)	80,7	MAIS PROVABLE	unfavourable

- **Thyroid cancer:** Já no §24 e no §94.

Lymphomas

Trait	Score (PGS Catalog)	Variants used	Percentile	Result	Judgement
Non-Hodgkin lymphoma	PGS000080 (Graff 2021, 19 variants)	18 19 (weighted coverage 96%)	1,1	MENOS PROVABLE	favourable
Diffuse lymphoma of large B cells	PGS002303 (Choi 2020, 9 variants)	8 9 (weighted coverage 90%)	5,4	MENOS PROVABLE	favourable
Follicular lymphoma	PGS002304 (Choi 2020, 6 variants)	6 6 (weighted coverage 100%)	0,3	MENOS PROVABLE	favourable
Hodgkin's lymphoma	PGS003449 (Berndt 2022, 21 variants)	16 21 (weighted coverage 68%)	0,8	MENOS PROVABLE	favourable

- **Non-Hodgkin lymphoma:** Já no §24.
- **Hodgkin's lymphoma:** Já no §24.

Leukemia and myeloma

Trait	Score (PGS Catalog)	Variants used	Percentile	Result	Judgement
Chronic lymphocytic leukemia	PGS000077 (Graff 2021, 75 variants)	69 75 (weighted coverage 89%)	68,9	TIPIC	typical
Chronic lymphocytic leukemia	PGS003453 (Berndt 2022, 43 variants)	38 43 (weighted coverage 87%)	78,4	TIPIC	typical
Acute lymphoblastic leukemia	PGS003448 (Berndt 2022, 15 variants)	13 15 (weighted coverage 89%)	73,1	TIPIC	typical
Multiple myeloma	PGS002306 (Choi 2020, 23 variants)	23 23 (weighted coverage 100%)	57,5	TIPIC	typical

- **Chronic lymphocytic leukemia:** Já no §24.
- **Multiple myeloma:** Já no §24.

Stomach and esophagus

Trait	Score (PGS Catalog)	Variants used	Percentile	Result	Judgement
Gastric cancer	PGS002299 (Choi 2020, 3 variants)	3 3 (weighted coverage 100%)	16,4	MENOS PROVABLE	favourable (low reliability)
Stomach cancer	PGS005161 (Zhu 2025, 12 variants)	6 12 (weighted coverage 36%)	6,3	MENOS PROVABLE	favourable (low reliability)
Oesophageal cancer	PGS002298 (Choi 2020, 14 variants)	13 14 (weighted coverage 98%)	84,8	MAIS PROVABLE	unfavourable
Oesophageal adenocarcinoma (genomic)	PGS003388 (Namba 2022, 357 thousand variants)	275.813 356.743 (weighted coverage 69%)	45,3	TIPIC	typical

- **Gastric cancer:** In §56 (Stomach).
- **Oesophageal cancer:** Já no §24.
- **Adenocarcinoma of the esophagus (genomic):** Já no §56.

Testis

Trait	Score (PGS Catalog)	Variants used	Percentile	Result	Judgement
Testis tumor	PGS000086 (Graff 2021, 52 variants)	48 52 (weighted coverage 90%)	23,9	TIPIC	typical
Testis tumor	PGS001164 (Tanigawa 2022, 280 variants)	184 280 (weighted coverage 72%)	7,9	MENOS PROVABLE	favourable

- **Testis tumor:** Já no §24 e no §66.

Brain

Trait	Score (PGS Catalog)	Variants used	Percentile	Result	Judgement
Glioma	PGS002302 (Choi 2020, 28 variants)	28 28 (weighted coverage 100%)	90,8	MAIS PROVABLE	unfavourable
Glioblastoma	PGS003384 (Namba 2022, 910 variants)	826 910 (weighted coverage 87%)	63,8	TIPIC	typical

- **Glioma:** Já no §24.

Non-melanoma skin

Trait	Score (PGS Catalog)	Variants used	Percentile	Result	Judgement
Basal cell carcinoma	PGS003417 (Seviiri 2022, 273 variants)	264 273 (weighted coverage 96%)	13,0	MENOS PROVABLE	favourable
Basal cell carcinoma	PGS000730 (Fontanyls 2021, 47 variants)	43 47 (weighted coverage 94%)	11,2	MENOS PROVABLE	favourable
Spinocellular carcinoma	PGS005209 (Gibson 2024, 20 variants)	20 20 (weighted coverage 100%)	3,4	MENOS PROVABLE	favourable
Spinocellular carcinoma	PGS000731 (Fontanyls 2021, 14 variants)	13 14 (weighted coverage 82%)	2,4	MENOS PROVABLE	favourable

- **Basal cell carcinoma:** Já no §24.
- **Spinocellular carcinoma:** Já no §24.

4. Key variants by topic

Loci of strong and replicated effect, grouped by organ. Each risk allele increases the risk by ~1.1–1.5× (relative). All genotypes were read in the chip, except ADH1B (imputation, DR2 = 0.88).

Lung

Nicotinic receptor (CHRNA5), telomerase (TERT), BRCA2 and HLA.

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
CHRNA5 (15q25)	rs16969968	D398N	G/A	chip	A (1 copy/s)	unfavourable (light)	D398N: allele A increases nicotine dependence and the risk of lung cancer (~1.3× per copy, especially in smokers)	Strong
BRCA2	rs11571833	K3326X	A/A	chip	T (0 copy/s)	favourable	K3326X: late truncation; ~2.5× more squamous lung cancer	Strong
TERT (5p15)	rs2736100		C/C	chip	G (2 copy/s)	unfavourable	TERT intron 2: lung adenocarcinoma (also in non-smokers) [GWAS Catalog: EGFR mutation-positive lung adenocarcinoma, allele G, OR 1.42, p=2E-31, PMID 27501781]	Strong
HLA (6p21)	rs3117582		T/T	chip	C (0 copy/s)	favourable	Region BAG6/APOM (6p21.33) of lung cancer [GWAS Catalog: Lung adenocarcinoma, C allele, OR 1.22, p=5E-12, PMID 19836008]	Moderate

Bexiga

Metabolization of aromatic (NAT2, UGT1A) and epithelial loci (PSCA, 8q24, FGFR3).

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
NAT2	rs1495741		A/A	chip	A (2 copy/s)	unfavourable	Slow acetylator tag: A/A : slow; ~1.3–1.5 plus bladder cancer with tobacco and aromatic amines	Strong
8q24 (MYC)	rs9642880		T/T	chip	T (2 copy/s)	unfavourable	8q24 near MYC: bladder cancer locus (~1.2× per copy) [GWAS Catalog: Bladder cancer, T allele, OR 1.24, p=4E-38, PMID 24163127]	Strong
PSCA	rs2294008		T/T	chip	T (2 copy/s)	unfavourable	The T allele increases the PSCA in the urothelium and stomach: bladder (~1,15×) and diffuse gastric (~1,3×)	Strong
TACC3/FGFR3	rs798766		C/C	chip	T (0 copy/s)	favourable	4p16 (FGFR3 neighbor) [GWAS Catalog: Bladder cancer, T allele, OR 1.22, p=7E-25, PMID 24163127]	Strong
UGT1A	rs11892031		A/A	chip	A (2 copy/s)	unfavourable	Carcinogenic glucuronidation (UGT1A) [GWAS Catalog: Bladder cancer, allele A, OR 1.23, p=1E-11, PMID 37210288]	Moderate

Pancreas

Blood group (ABO), pancreatic development (NR5A2) and CLPTM1L/TERT.

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
ABO	rs505922		T/T	chip	C (0 copy/s)	favourable	The C allele marks the groups A/B (non-O): ~1.2–1.3× more pancreatic cancer	Strong

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
NR5A2	rs3790844	A/A	chip		A (2 copy/s)	unfavourable	Nuclear receptor LRH-1 (exocrine pancreas) [GWAS Catalog: Pancreatic cancer, allele A, OR 1.23, p=8E-16, PMID 29422604]	Strong
CLPTM1L/TERT	rs401681	C/T	chip		T (1 copy/s)	unfavourable (light)	5p15.33 (CLPTM1L/TERT): T allele is associated with ~1.19× more pancreatic cancer; in the same locus, C is associated with more bladder and lung cancer (pleiotropic effect) [GWAS Catalog: Pancreatic cancer, T allele, OR 1.19]	Moderate
KLF5	rs9573163	C/G	chip		G (1 copy/s)	unfavourable (light)	13q22.1 (KLF5): pancreatic cancer [GWAS Catalog: Pancreatic cancer, G allele, OR 1.26, p=5E-13, PMID 22158540]	Moderate

Rim

Hypoxia and cell cycle loci of renal cell carcinoma.

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
11q13.3 (CCND1)	rs7105934	A/A	chip		G (0 copy/s)	favourable	Enhancer of CCND1 linked to HIF; the minor allele protects [GWAS Catalog: Malignant neoplasm of kidney, except for pheCode 189.11, allele G, beta 0.270 [0.2-0.34] unit increase, p=8E-25, PMID 39024449]	Strong
SCARB1	rs4765623	C/C	chip		T (0 copy/s)	favourable	HDL SR-BI receiver: renal cell carcinoma [GWAS Catalog: Clear renal cell carcinoma, allele T, OR 1.15, p=3E-23, PMID 38671320]	Moderate

Tyroid

The two strongest loci of papillary cancer (detail in §108).

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
FOXE1 (9q22)	rs965513	A/G	chip		A (1 copy/s)	unfavourable (light)	~1.7× more papillary cancer of the thyroid per copy	Strong
NKX2-1/MBIP	rs944289	C/T	chip		T (1 copy/s)	unfavourable (light)	~1.3× more papillary cancer of the thyroid per copy	Strong

Lymphomas and leukemias

Lymphoid transcription factors (IRF4, ARID5B, IKZF1) and HLA class II.

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
IRF4	rs872071	A/G	chip		G (1 copy/s)	unfavourable (light)	IRF4 (6p25): chronic lymphocytic leukemia and Hodgkin's lymphoma [GWAS Catalog: Chronic lymphocytic leukemia, allele G, OR 1.54, p=2E-20, PMID 18758461]	Strong
ARID5B	rs7089424	T/G	chip		T (1 copy/s)	unfavourable (light)	ARID5B: acute lymphoblastic leukemia (~1.6× per copy) [GWAS Catalog: Acute lymphoblastic leukemia (childhood), T allele, OR 1.64, p=2E-62, PMID 29348612]	Strong

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
IKZF1	rs4132601	T/G	chip		C (1 copy/s)	unfavourable (light)	Ikaros (lymphoid development): acute lymphoblastic leukemia [GWAS Catalog: Acute lymphoblastic leukemia (childhood), C allele, OR 1.69, p=1E-19, PMID 19684604]	Strong
HLA class II	rs10484561	T/T	chip		G (0 copy/s)	favourable	HLA-DQB1: follicular lymphoma [GWAS Catalog: Follicular lymphoma, G allele, OR 1.95, p=1E-29, PMID 20639881]	Strong

Stomach and esophagus

Gastric mucus (MUC1, PLCE1), esophageal FOXF1 and ADH1B (acetaldehyde).

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
MUC1 (1q22)	rs4072037	C/C	chip		T (0 copy/s)	favourable	Splicing of MUC1 (gastric mucus barrier): gastric cancer [GWAS Catalog: Non-cardia gastric cancer, allele T, OR 1.35, p=6E-17, PMID 26701879]	Strong
PLCE1	rs2274223	A/A	chip		G (0 copy/s)	favourable	PLCE1: squamous cell carcinoma of the esophagus and gastric cardia [GWAS Catalog: Esophageal cancer, G allele, OR 1.34, p=4E-20, PMID 21642993]	Moderate
FOXF1	rs9936833	C/T	chip		C (1 copy/s)	unfavourable (light)	FOXF1 (16q24): Barrett and adenocarcinoma of the esophagus [GWAS Catalog: Barrett's esophagus, allele C, OR 1.14, p=3E-10, PMID 22961001]	Moderate
ADH1B	rs1229984	C/C	imputation DR2=0.88		C (2 copy/s)	unfavourable	ADH1B*2: alcohol metabolism; relevant only with alcohol [GWAS Catalog: Esophageal cancer, C allele, beta 0.712 [0.61-0.81] unit increase, p=2E-46, PMID 34594039]	Moderate

Testis

Via KIT-ligand (KITLG, SPRY4) and apoptosis (BAK1).

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
KITLG	rs995030	A/A	chip		G (0 copy/s)	favourable	KITLG (KIT pathway of germ cells): the G allele gives ~2.5× per copy	Strong
SPRY4	rs4624820	G/A	chip		G (1 copy/s)	unfavourable (light)	SPRY4: germ cell tumor [GWAS Catalog: Testicular germ cell tumor, allele G, OR 1.47, p=3E-57, PMID 28604728]	Strong
BAK1	rs210138	A/G	chip		G (1 copy/s)	unfavourable (light)	BAK1 (apoptosis): testis tumor [GWAS Catalog: Testicular germ cell tumor, allele G, OR 1.48, p=4E-37, PMID 28604728]	Moderate

Brain (glioma)

Telomeres (TERT, RTEL1), CDKN2B-AS1, CCDC26 and PHLDB1.

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
TERT	rs10069690		T/T	chip	T (2 copy/s)	unfavourable	Intron 4 of TERT: glioma (~1.3× per copy)	Strong
RTEL1	rs6010620		G/G	chip	G (2 copy/s)	unfavourable	Helicase of the telomeres RTEL1: glioma (~1.3× per copy)	Strong
CDKN2B-AS1 (9p21)	rs4977756		A/A	chip	G (0 copy/s)	favourable	9p21 (CDKN2A/B): glioma [GWAS Catalog: Glioma, G allele, OR 1.24, p=7E-15, PMID 19578367]	Strong
CCDC26 (8q24)	rs55705857		A/A	chip	G (0 copy/s)	favourable	8q24.21: oligodendroglioma and astrocytoma IDH-mutated (~6× per copy, rare allele) [GWAS Catalog: Non-glioblastoma glioma, allele G, OR 3.39, p=7E-149, PMID 28346443]	Strong
PHLDB1	rs498872		G/G	chip	A (0 copy/s)	favourable	11q23.3: glioma IDH-mutated [GWAS Catalog: Glioma, allele A, OR 1.25, p=3E-9, PMID 30714141]	Moderate

Non-melanoma skin

Pigmentation (MC1R, IRF4, HERC2, SLC45A2) and TP53.

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
MC1R	rs1805007	R151C	C/C	chip	T (0 copy/s)	favourable	R151C: light skin and little tolerance to the sun ~1.5-2+ basal cell carcinoma and squamous cell carcinoma	Strong
MC1R	rs1805008	R160W	C/C	chip	T (0 copy/s)	favourable	R160W: variant "r" of MC1R	Strong
MC1R	rs1805009	D294H	G/G	chip	C (0 copy/s)	favourable	D294H: strong "R" variant of MC1R	Strong
IRF4	rs12203592		C/C	chip	T (0 copy/s)	favourable	The T allele reduces the IRF4 in melanocytes: freckles, sensitivity to sun and squamous cell carcinoma	Strong
HERC2/OCA2	rs12913832		A/G	chip	G (1 copy/s)	unfavourable (light)	The G allele (2 copies = clear eyes) is associated with less pigment and more skin cancer	Moderate
SLC45A2	rs16891982	L374F	C/G	chip	G (1 copy/s)	unfavourable (light)	L374F: the G (Phe374) allele gives clear pigmentation	Moderate
TP53 (3' polyA)	rs78378222		T/T	chip	C (0 copy/s)	favourable	Rare variant of the polyadenylation signal of TP53: ~2× plus basal cell carcinoma and glioma	Strong

5. Favourable

- **Lymphomas:** non-Hodgkin 1.1, diffuse of large B cells 5.4, follicular 0.3, Hodgkin 0.8; HLA class II rs10484561 T/T. **Moderate.**
- **Rim:** percentiles 0.8 and 22.4 (107 variants); 11q13.3/CCND1 A/A and SCARB1 C/C. **Moderate.**
- **Non-melanoma skin:** basal cell 13.0 and 11.2; squamous cell 3.4 and 2.4; MC1R R151C, R160W and D294H absent, IRF4 C/C. **Moderate.**
- **Testis:** 23.9 and 7.9; KITLG rs995030 A/A (without the strongest risk allele, $\sim 2.5\times$ per copy). **Moderate.**
- **Stomach:** 16.4 and 6.3 (36% coverage); MUC1 C/C, PLCE1 A/A. Weak (small scores).
- **Normal rare variants:** BRCA2 K3326X A/A, TP53 rs78378222 T/T, CHEK2 I157T (S108). Strong for the tested variants.
- **Pancreas:** ABO rs505922 T/T (Probable O group, $\sim 0.8 \times$ risk). **Moderate.**

6. Unfavourable

- **Bexiga:** percentiles 94.9, 90.4 and 86.1 (genomic in 57.2); NAT2 slow A/A, PSCA T/T, 8q24 T/T and UGT1A A/A (2 copies each). Moderate to strong (NAT2 slow × aromatic amines).
- **Glioma:** percentile 90.8 (glioblastoma in 63.8); TERT rs10069690 T/T and RTEL1 G/G. Very low absolute risk (about 0.5% in life). **Moderate.**
- **Pancreas:** 83.1 (22 variants; typical genomics in 45.0); NR5A2 A/A, 1 allele in CLPTM1L and KLF5. **Weak to moderate.**
- **Esophagus:** 84.8 (14 variants; typical genomic adenocarcinoma in 45.3); FOXF1 1 allele; ADH1B C/C (slow metabolism of alcohol, irrelevant with zero alcohol). **Weak.**
- **Tyroid:** 79.1 and 80.7; FOXE1 and NKX2-1 with 1 allele (see §108). **Moderate.**
- **Leukemia:** LLC 68.9 and 78.4, LLA 73.1 (typical, above the median); IRF4, ARID5B and IKZF1 with 1 allele. **Weak.**
- **Lung (only with tobacco):** CHRNA5 G/A and TERT rs2736100 with 2 alleles; typical scores. Moderate for smokers; no practical coefficients without tobacco.

7. Mechanisms of ageing

Mapping in 84 mechanisms of aging of Sérgio F immortality objectives (only the mechanisms affected by this report).

Mechanism	Genetic finding	Meaning
Somatic Mutation Accumulation	All these cancers are born from somatic mutations; the polygenic risk decides the rate at which the risk clones accumulate in the bladder, brain and thyroid	Unfavourable
Hayflick's Telomeres and Limit	TERT rs10069690 T/T, TERT rs2736100 and RTEL1 G/G: longer telomeres, less senescence, more glioma and thyroid	Conditional
DNA Damage and DNA Repair Deficit	Slow NAT2: more aromatic amines in the urothelium; BRCA2 K3326X and TP53 normal	Unfavourable (mild)
Genomic instability	No variants of high effect tested; polygenic risk	Neutral
Immunosenescence	Very unlikely lymphomas; the immune surveillance of clones remains	Favourable
Chronic Inflammation	Reflux, Helicobacter pylori and chronic urinary infections are inflammatory accelerators in the esophagus, stomach and bladder	Conditional
Clonal hematopoiesis (CHIP)	LLC above the median; clonal hematopoiesis precedes several blood neoplasms	Conditional

8. Practical suggestions

All suggestions are suggestion (not applied), consistent with the current genetic plan, and should be discussed with the doctor.

1. **Bexiga:** to see a urology with any blood in the urine, even if it is a single episode and without pain. It is the earliest signal, and cytology screening is not recommended. Keeping zero tobacco, drinking plenty of water and avoiding professional exposure to aromatic amines and paints. Avoid much burned meat (heterocyclic amines with slow NAT2). *Suggestion (not applied).*
2. **Stomach:** Helicobacter pylori test (breathing or feces) once and eradication if positive. It is the measure of greatest impact on gastric cancer, and current standards recommend it even without symptoms. *Suggestion (not applied).*
3. **Esophagus:** if there is heartburn or reflux for more than 5 years, discuss an endoscopy with the doctor (Barrett's esophagus). Avoid very hot drinks ($> 65^{\circ}\text{C}$). Zero alcohol and zero tobacco already protect. *Suggestion (not applied).*
4. **Pancreas:** there is no screening outside of high-risk families. The warning signs are recent onset diabetes after the 50 years without explanation, jaundice, persistent abdominal pain that radiates to the back and unexplained weight loss. If there are first-degree 2 relatives with pancreatic cancer, ask for a surveillance program. *Suggestion (not applied).*
5. **Brain:** there is no screening. Saving the head to ionizing radiation (TC only when indicated) and valuing new and progressive headaches, de novo seizures or neurological deficits. *Suggestion (not applied).*
6. **Blood:** annual blood count (it is already part of the routine analysis). A persistent lymphocytosis justifies immunophenotyping (monoclonal B lymphocytosis or LLC). *Suggestion (not applied).*
7. **Skin:** annual skin observation by dermatologist and sun protection. The risk of basocellular and squamous cells is low, but the age and the accumulated sun weigh more than the genes. *Suggestion (not applied).*
8. **Testis and thyroid:** optional testis self-examination; thyroid according to §108. *Suggestion (not applied).*
9. **Common to all:** fiber (the more, the better), low LDL and ApoB, exercise, zero alcohol and zero tobacco, ferritin below 100 and vitamin D at optimal levels, as already in the plan. *Suggestion (not applied).*

9. Limitations

- The small scores (3–28 variants) reflect mainly the strong loci and are unstable. When the genomic score of the same organ is typical (pancreas, bladder, esophagus), the real risk is probably moderate.
- Some scores (stomach, genomic esophagus, 280 testis variants) have partial coverage (36–72%) and were developed in other populations; they read with caution.
- The chip does not read the high-effect genes completely (BRCA1/2, Lynch, VHL, CDH1, CDKN2A, TP53). Family history remains essential.
- The absolute risks of these cancers are mostly low (glioma, testis, lymphomas); a high percentile does not mean high risk in absolute terms.
- Conclusions only from genes; it is not a diagnosis.

10. Methods and sources

- **Genotypes:** Sérgio F chip (GRCh38, strand +). The absent SNPs come from local imputation (Beagle 5.5, public panel of complete genomes to 30x), used only with $DR2 \geq 0.8$. No genetic data came out of the analysis computer.
- **Alleles and strand:** The published effect allele is compared with the alleles of the Imposed VCF. The alleles reported in the strand – were converted (4 SNPs) and in the palindromics the strand + (GWAS Catalog standard and PGS Catalog) was assumed.
- **Polygenic scores:** GRCh38 harmonized files from PGS Catalog (33 scores in 13 organs or types of cancer). Score = Δ coefficient $\times$ dosage. The percentile is empirical in the reference population ($n = 633$), with the same variants.
- **Interpretation:** Graff 2021 and Choi 2020 (multi-cancer scores), Hu 2024 (genomic scores), Kachuri 2020 and Kothers 2023 (bexiga), Berndt 2022 (lymphoid neoplasms), Purdue 2024 (rim), Namba 2022 (esophageal and glioblastoma), Fontanillas 2021 and Seviiri 2022 (non-melanoma skin), Rothman 2010 (NAT2), Malfertheiner 2022 (Helicobacter pylori), GWAS Catalog and ClinVar.

11. Annex: all variants analysed

Topic	Gene	SNP	Genotype	Source	Effect allele	Copies	Chain	Interpretation
Lung	CHRNA5 (15q25)	rs16969968	G/A	chip	A	1	strand +	1 copy of risk allele
Lung	BRCA2	rs11571833	A/A	chip	T	0	strand + (palindromic; assumed +)	without the risk allele
Lung	TERT (5p15)	rs2736100	C/C	chip	G	2	complement (strand)	2 risk allele copies
Lung	HLA (6p21)	rs3117582	T/T	chip	C	0	complement (strand)	without the risk allele
Bexiga	NAT2	rs1495741	A/A	chip	A	2	strand +	2 risk allele copies
Bexiga	8q24 (MYC)	rs9642880	T/T	chip	T	2	strand +	2 risk allele copies
Bexiga	PSCA	rs2294008	T/T	chip	T	2	strand +	2 risk allele copies
Bexiga	TACC3/FGFR3	rs798766	C/C	chip	T	0	strand +	without the risk allele
Bexiga	UGT1A	rs11892031	A/A	chip	A	2	strand +	2 risk allele copies
Pancreas	ABO	rs505922	T/T	chip	C	0	strand +	without the risk allele
Pancreas	NR5A2	rs3790844	A/A	chip	A	2	strand +	2 risk allele copies
Pancreas	CLPTM1L/TERT	rs401681	C/T	chip	T	1	strand +	1 copy of risk allele
Pancreas	KLF5	rs9573163	C/G	chip	G	1	strand + (palindromic; assumed +)	1 copy of risk allele
Rim	11q13.3 (CCND1)	rs7105934	A/A	chip	G	0	strand +	without the risk allele
Rim	SCARB1	rs4765623	C/C	chip	T	0	strand +	without the risk allele
Tyroid	FOXE1 (9q22)	rs965513	A/G	chip	A	1	strand +	1 copy of risk allele
Tyroid	NKX2-1/MBIP	rs944289	C/T	chip	T	1	strand +	1 copy of risk allele
Lymphomas and leukemias	IRF4	rs872071	A/G	chip	G	1	strand +	1 copy of risk allele
Lymphomas and leukemias	ARID5B	rs7089424	T/G	chip	T	1	strand +	1 copy of risk allele
Lymphomas and leukemias	IKZF1	rs4132601	T/G	chip	C	1	complement (strand)	1 copy of risk allele
Lymphomas and leukemias	HLA class II	rs10484561	T/T	chip	G	0	strand +	without the risk allele
Stomach and esophagus	MUC1 (1q22)	rs4072037	C/C	chip	T	0	strand +	without the risk allele
Stomach and esophagus	PLCE1	rs2274223	A/A	chip	G	0	strand +	without the risk allele
Stomach and esophagus	FOXF1	rs9936833	C/T	chip	C	1	strand +	1 copy of risk allele
Stomach and esophagus	ADH1B	rs1229984	C/C	imputation DR2=0.88	C	2	strand +	2 risk allele copies
Testis	KITLG	rs995030	A/A	chip	G	0	strand +	without the risk allele

Topic	Gene	SNP	Genotype	Source	Effect allele	Copies	Chain	Interpretation
Testis	SPRY4	rs4624820	G/A	chip	G	1	strand +	1 copy of risk allele
Testis	BAK1	rs210138	A/G	chip	G	1	strand +	1 copy of risk allele
Brain (glioma)	TERT	rs10069690	T/T	chip	T	2	strand +	2 risk allele copies
Brain (glioma)	RTEL1	rs6010620	G/G	chip	G	2	strand +	2 risk allele copies
Brain (glioma)	CDKN2B-AS1 (9p21)	rs4977756	A/A	chip	G	0	strand +	without the risk allele
Brain (glioma)	CCDC26 (8q24)	rs55705857	A/A	chip	G	0	strand +	without the risk allele
Brain (glioma)	PHLDB1	rs498872	G/G	chip	A	0	strand +	without the risk allele
Non-melanoma skin	MC1R	rs1805007	C/C	chip	T	0	strand +	without the risk allele
Non-melanoma skin	MC1R	rs1805008	C/C	chip	T	0	strand +	without the risk allele
Non-melanoma skin	MC1R	rs1805009	G/G	chip	C	0	strand + (palindromic; assumed +)	without the risk allele
Non-melanoma skin	IRF4	rs12203592	C/C	chip	T	0	strand +	without the risk allele
Non-melanoma skin	HERC2/OCA2	rs12913832	A/G	chip	G	1	strand +	1 copy of risk allele
Non-melanoma skin	SLC45A2	rs16891982	C/G	chip	G	1	strand + (palindromic; assumed +)	1 copy of risk allele
Non-melanoma skin	TP53 (3' polyA)	rs78378222	T/T	chip	C	0	complement (strand)	without the risk allele